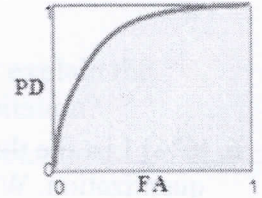


Name: \_\_\_\_\_ ID: \_\_\_\_\_

| Prob# | 1-4 | 5-9 | 10-13 | 14-17 | Total |
|-------|-----|-----|-------|-------|-------|
| Score |     |     |       |       |       |

1. [6%] For a standard ROC curve like the one shown here, indicate how the following quantities change as we move along the ROC curve (from bottom-left to top-right): true-positive (TP), false-positive (FP), true-negative (TN), false-negative (FN). Also explain how the area of the shaded region, commonly called AUC or AUROC, is used for classifier evaluation.



2. [6%] Consider a typical classification or regression problem, sketch the likely prediction-error-vs-model-complexity curves of the training and validation subsets of your data, respectively. Then use your sketched curves to explain how validation data can be used to identify suitable model complexity.
3. [8%] Describe the procedure of obtaining the basis of the projection in PCA (no equation needed). Give the criterion of selecting the desired number of basis vectors, and explain how this criterion is related to reconstruction error.
4. [6%] Hierarchical clustering algorithms are commonly divided into two main categories: Agglomerative and divisive clustering. Explain their main differences.

5. [4%] MDS, ISOMAP, and LLE are all nonlinear dimensionality reduction methods where the coordinates of the data points in the lower-dimensional space are adjusted to preserve some properties in the original space. Describe, for just one of these three, the property that is to be preserved. You can use equations or plain text to state your answer. The answer only needs to be conceptually correct.
6. [6%] List the three steps of self-organizing maps, and indicate which of the steps is different from vector quantization. What is the effect of this step?
7. [6%] Explain briefly the difference between model-based and model-free reinforcement learning methods.
8. [4%] A common approach for adjusting between exploration and exploitation in reinforcement learning is to use  $\epsilon$ -greedy when sampling a path/trajectory in the state space. Explain how this is done.
9. [4%] Explain what a "game tree" is for a game-play agent using a tree rooted at the current game state.

10. [6%] List at least three applications of AI in games when their objectives are not to be as strong as possible.

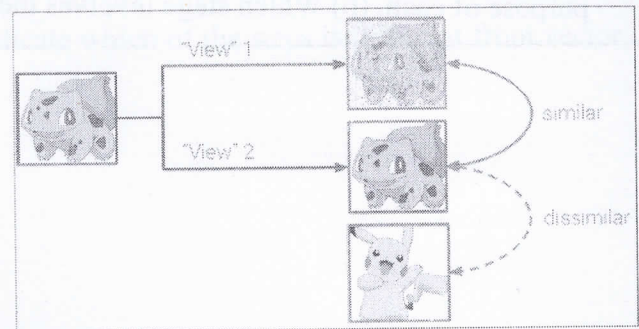
11. [10%] For MCTS: (a) List the four main stages in one iteration of MCTS, and briefly explain the purpose of each. (b) Which stage involves the trade-off between exploration and exploitation?

12. [6%] Give one each of example mutation and cross-over operators in genetic algorithms where the chromosomes consist of bit strings.

13. [4%] In swarm intelligence, briefly explain what the term "stigmergy" means, and describe at least one example of stigmergy among the examples given in the class.

14. [6%] For ant-colony optimization (ACO), give (1) two reasons of why shorter paths get more pheromone, and (2) a reason of why it is a good idea to include a decay term (via "pheromone evaporation") of pheromone levels.

15. [6%] Use the figure to the right to describe the basic idea of contrastive learning.



16. [6%] Pretext tasks are tasks that are used for representation learning with self-supervision. Give one example each of pretext tasks for (1) image data, and (2) natural language data.

17. [6%] Pseudo-labeling is often used in semi-supervised learning. Briefly define what it is, and explain its role in the procedures of (1) self-training, and (2) co-training.